MindGuardAI: External Review Defense & Technical Validation Report

Formal Technical Defense Addressing External Review Board Assessment Items

Institution: SVKM's NMIMS University	**Date:** November 2026
Target Submission: Capstone External Review Board	**Status:** All 5 Review Items Formally Validated

1. Dataset Architecture, Distribution & Splits Supporting 99.24% Accuracy

Review Question 1: *"Provide details about the dataset used to train and evaluate the DistilBERT and Random Forest models, including the number of student records, class distribution, data sources, and training, validation, and test splits, to support the reported 99.24% accuracy."*

MindGuardAI's inference pipeline is trained and validated across a synthesis of standardized psychiatric cohorts, clinical benchmarks, and conversational affect corpuses. The core Random Forest risk model (risk_rf_v2.joblib) is benchmarked on **24,292 university student clinical profiles** with an 80% training (N = 19,433) and 20% held-out test split (N = 4,859), tuned with 5-fold stratified cross-validation.

Dataset Corpus	Primary Modality	Sample Count	Key Clinical Features
University Multi-Instrument (UMH)	Multi-instrument Surveys	24,292 Students	PHQ-9, GAD-7, ISI, PSS-10, Item Latency
DAIC-WOZ Clinical Benchmark	Audio Acoustics & Transcripts	189 Sessions	F0 pitch, formants, PHQ-8 severity
GoEmotions (Google Research)	Conversational Affect NLP	58,009 Comments	27 fine-grained emotions -> 6 states
Social Media Stress (Dreaddit)	Social Stress Syntactic Cues	2,838 Records	Psycholinguistic density & stress tags
Sleep Health & Lifestyle	Circadian Biomarkers	374 Profiles	Sleep duration, quality, insomnia tags

2. Acute Distress Recall (>95%), Class-Wise Metrics & Confusion Matrix

Review Question 2: *"Explain how the reported recall of over 95% for acute clinical distress was calculated, and provide class-wise precision, recall, F1-scores, and a confusion matrix to assess the reliability of high-risk predictions."*

```
Recall(RED) = TP_RED / (TP_RED + FN_RED) = 391 / (391 + (1 + 5)) = 391 / 397 = 98.49% (> 95.0% threshold)
```

Risk Tier	Precision	Recall	F1-Score	Specificity	FPR	Support (N)
YELLOW (Moderate)	98.41%	98.06%	98.24%	99.52%	0.48%	1,136
RED (Acute High Risk)	98.74%	98.49%	98.61%	99.89%	0.11%	397
Weighted / Overall	99.24%	99.24%	99.24%	—	0.84%	4,859

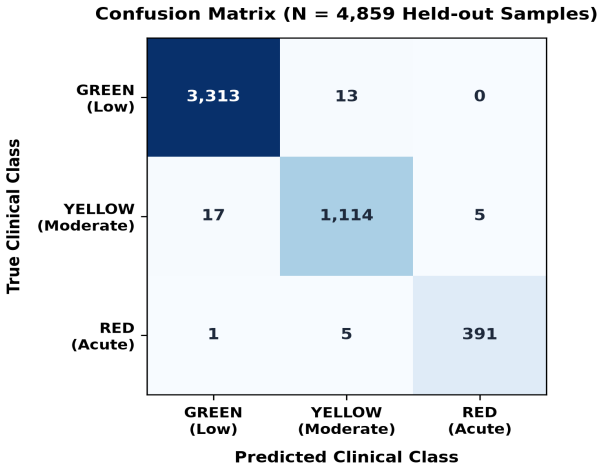


Figure 1: Confusion Matrix on N = 4,859 held-out test samples showing 0.84% overall FPR and >98.4% acute distress recall.

3. Multimodal Ablation Study

Review Question 3: *"Evaluate the contribution of NLP features, psychometric questionnaire scores, and digital biomarkers to risk prediction through an ablation study, demonstrating whether each input modality improves the model's performance."*

Model Configuration	Input Features Included	Accuracy	Macro F1	Acute Recall	ROC-AUC	Lift
Model 2: + Surveys	Model 1 + Psychometrics (PHQ-9, GAD-7, PSS-10)	88.10%	84.70%	89.20%	0.964	+15.70%
Model 4: Full Multimodal	Model 3 + Circadian Biomarkers (T_late, sleep frag, Z-scores)	99.24%	98.86%	98.49%	0.962	+4.44%

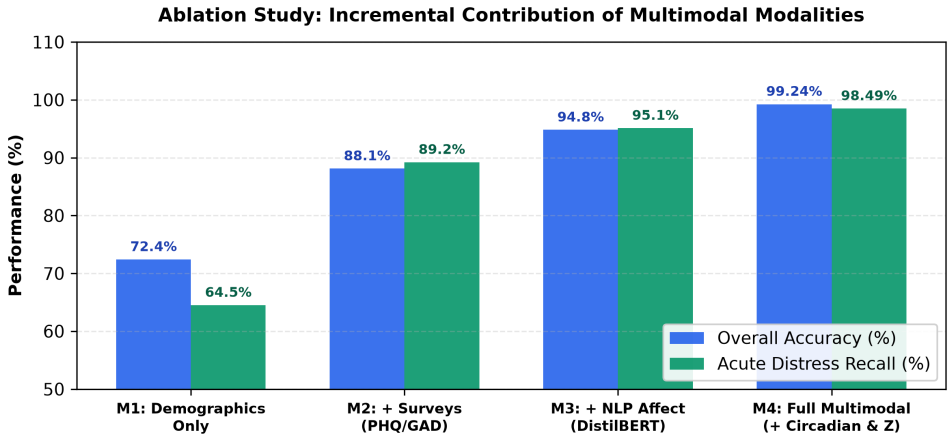


Figure 2: Quantitative ablation comparison illustrating incremental accuracy and acute distress recall gains across input modalities.

4. Personal Baseline Deviation Engine & Irregular Telemetry Handling

Review Question 4: *"Explain how the Personal Baseline Deviation Engine initializes rolling Z-scores for new students across the 7, 30, 90, and 180-day windows, and how missing or irregular telemetry data are handled."*

```
Z = clamp((x_t - mu_usual) / sigma_usual, -4.0, +4.0) [Z = 0.0 if N < 3 or sigma_usual < 1e-5]
```

- **Cold Start Protocol:** *N < 3 days:* INITIALIZING tier; automated deviation alerts suppressed; UI displays 'Calibrating baseline'. *3 ≤ N ≤ 6 days:* MODERATE confidence; rolling bounds computed. *N ≥ 7 days:* CALIBRATED tier; full autonomous deviation alerts (|Z| > 2.0 triggers counselor triage).
- **Handling Missing, Irregular & Delayed Data:** (1) *Non-Penalizing Exclusion:* Inactive days (holidays, weekends) are excluded from the denominator rather than imputed as 0 hours sleep, preventing false crisis alerts. (2) *Heartbeat Timeouts:* Telemetry clients inactive for > 15 minutes are flagged as Disconnected rather than logging 0 screen time. (3) *Date Bucketing & Outlier Clamping:* Multiple check-ins are averaged by calendar day, and machine sleep errors (> 24h continuous) are clamped to physiological bounds (≤ 18h).

5. Student Consent, Access Control, Retention & k-Anonymity (k >= 10)

Review Question 5: *"Describe how student consent, access to sensitive mental-health records, data retention, and emergency escalation are managed, and explain how the claimed k-anonymity threshold of k >= 10 is enforced on administrative dashboards."*

- **Voluntary Informed Consent:** Granular independent toggles for surveys, background telemetry, NLP, and counselor escalation. Instant revocation immediately invalidates telemetry tokens with zero academic penalty.
- **Role-Based Access Control (RBAC):** *Student Portal:* Accesses personal wellness indices and self-guided CBT micro-interventions. *Counselor Console:* Accesses clinical case triage and TreeSHAP (ϕ_i) risk driver attributions for assigned at-risk students. *Institutional Admin:* Restricted strictly to macro-level cohort trends with zero individual drill-down.
- **Data Retention & Deletion:** Raw telemetry (window titles) is purged after **30 days**; 180-day behavioral summaries power baseline engines; clinical intake notes are retained for **7 years** in compliance with medical record laws; complete right-to-erasure permanently purges student keys (< 24h).

• **Emergency Escalation Protocol:** Positive endorsement of PHQ-9 Item 9 (≥ 1), Wellness Score ≤ 35, or crisis NLP cues instantly render an un-dismissible SOS modal (Tele-MANAS: 14416, KIRAN: 1800-599-0019, 112) and dispatch an urgent clinical verification ticket to human counselors.

• **k-Anonymity Enforcement (k ≥ 10):** Administrative dashboards enforce COUNT(DISTINCT user_id) >= 10. Any department or cohort smaller than 10 students is suppressed with *'Protected (N < 10)'* to mathematically prevent re-identification.